

CARLYN LEE

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Accomplished Software Engineer specializing in scalable solutions, system reproducibility, and advanced visualization tools for high-performance computing. Experienced in interdisciplinary collaboration, complex algorithms, and system optimization with: *C/C++, Python, Java, MATLAB, Ansys STK*

EXPERIENCE

Oligo Space

Jan 2025 - Present

Head of Software Engineering

- Architected Zenith, an end-to-end spacecraft design automation framework connecting payload requirements to complete satellite configurations.
- Developed automated pipelines for S-band, X-band, and UHF link budget analysis, regulatory filings (FCC Part 25 / ITU Appendix B & C), and integration with commercial ground-station networks.
- Integrated onboard networking components (SatCat5 FPGA switch, Arty A7-100T, Raspberry Pi-based data paths) for payload developer kits and flight-software interfaces.
- Designed internal tooling to support rapid spacecraft iteration, GMAT-based orbit analysis, and reinforcement-learning prototypes for structural mass-model optimization.

NASA Jet Propulsion Laboratory, California Institute of Technology Aug 2012 - Nov 2024
Applications Software Engineer Pasadena, CA

- **Tools Development:** Built deep space mission telecom simulation and modeling tools. Led and mentored remote interns on mission-critical systems. Automated operations enabling three Deep Space Network sites to manage the network during day shifts instead of 24/7.
- **Mission Operations:** Perseverance Rover telecom chair, responsible for rover communications health using data from Mars. Delivered tools reducing operation response-time from several hours to 20 minutes.
- **Supercomputing:** Developed Lunar terrain mask algorithms for Shaheen II supercomputer, enabling 10m-resolution link coverage maps. Parallelized Deep Space Optical Communications simulations, achieving bit error rate of 10e-8, a 1000x improvement.
- **Fieldwork:** Reliable multi-agent data transfer processes at transport layer. Field tested autonomous systems in maritime and subterranean environments, contributing to 1st place in DARPA's Urban Circuit Challenge.
- **Verification & Validation:** Developed automated tools and test procedures for avionics flight software, executing tests on flight hardware.

EXTRACURRICULAR, VOLUNTEER & PROFESSIONAL AFFILIATIONS

2019 - Present Global Network Advancement (GNA-G) Data Intensive Sciences Working Group, enabling high-throughput data transfers for Caltech, SCLinet demonstration at Supercomputing 24-25.

2018 - Present SoCal Linux Expo Volunteer, deploying on-site network infrastructure and AV systems.

2014 - Present Interplanetary Small Satellite Conference committee member, managing budgets and logistics for 100+ industry leaders.

EDUCATION

California State University, Fullerton

M.S. Computer Science

August 2012

B.S. Computer Science, Minor in Mathematics

July 2011

PUBLICATIONS

- K. Cheung; V. Vilnrotter; M. Sanchez-Net; C. Lee. High Data Rates from the Outer Solar System. IEEE Aerospace Conference. Big Sky, MT, USA. March 2025.
- Vander Hook, J. V., Seto, W., Nguyen, V., Hasnain, Z., Lee, C.-A., Gallagher, L., Halpin-Chan, T., Varahamurthy, V., & Angulo, M. (2022). Swarms of Pirates: Red Team Exercises Using Autonomous High-Speed Maneuvering Surface Vessels. *Field Robotics*, 2, 872–909.
- C. Lee, M. Shaikh, C.H. Lee and D. Michels. Lunar Terrain Coverage Analysis Data Delivery Workflow, AIAA 2021-4039. ASCEND 2021. November 2021.
- A. Agha et al.. NeBula: Quest for Robotic Autonomy in Challenging Environments; TEAM CoSTAR at the DARPA Subterranean Challenge. *Journal of Field Robotics*, 2021.
- C. Lee, H. Xie, C.H. Lee, D. Lyakhov, and D. Michels. In Silico Methods for Space System Analysis: Optical Link Coding Performance and Lunar Terrain Masks. In *AIAA ASCEND*, Las Vegas, NV, 16-18 Nov. 2020.
- D. Abraham, B. MacNeal, D. Heckman, Y. Chen, J. Wu, K. Tran, A. Kwok and C. Lee. Recommendations Emerging from an Analysis of NASA’s Deep Space Communications Capacity. In *International Conference On Space Operations (SpaceOps 2018)*, Marseille, France, May 2018.
- J. Lad, M. Johnston, D. Tran, D. Brown, K. Roffo, and C. Lee. Complexity-Based Link Assignment for NASA’s Deep Space Network for Follow-the-Sun Operations. In *International Conference On Space Operations (SpaceOps 2018)*, Marseille, France, May 2018.
- K. Pinover, M. Johnston, C. Lee. Optimizing SmallSat Scheduling for NASA’s Deep Space Network. In *International Workshop on Planning and Scheduling for Space (IWPSS 2017)*. Pittsburgh, PA, June 2017.
- D. Morabito, D. Kahan, K. Oudrhiri, and C. Lee. Cassini Downlink Ka-Band Carrier Signal Analysis. *The Interplanetary Network Progress Report*, Volume 42-208, February 15, 2017.
- K. Cheung, D. Abraham, M. Sanchez-Net, K. Tran, C. Lee. Traffic modeling for Deep Space Network in the Human Exploration Era. In *SpaceOps 2016 Conference*, Daejeon, Korea, May 16-20, 2016.
- M. Johnston, C. Lee, C. Lau, K. Cheung, M. Levesque, B. Carruth, A. Coffman, M. Wallace. Integrating space communication network capabilities via web portal technologies. In *SpaceOps 2014 13th International Conference on Space Operations*, Pasadena, California, May 5-9, 2014.
- C. Lee, C.H. Lee. Cancer Screening Using Multi-modal Differential Principal Orthogonal Decomposition. In *2013 13th International Conference on Computational Science and Its Applications*, Ho Chi Minh City, Vietnam, June 24-27 2013.
- C. Lee, C.H. Lee. Extended Principal Orthogonal Decomposition Method for Cancer Screening. *International Journal of Bioscience, Biochemistry and Bioinformatics* vol. 2, no. 2, pp. 136-141, 2012.
- C. Lee. Rest architecture for link analysis tools portal. NASA Undergraduate Student Research Program (USRP), Pasadena, California, August 2011.